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Department of Computer Science & Engineering

LABORATORY FILE 4

Subject / Course	Full Stack Web Development
Experiment No.	04
Experiment Title	Node.js – Modules, FS, HTTP, Streams & npm
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Experiment No. 04

Title: Node.js – Custom Modules, File System, HTTP Server, Streams & npm

Aim

To learn and implement core Node.js features including custom modules, file and directory operations, path and OS modules, EventEmitter, HTTP server with manual routing and query parsing, streams, npm packages, command-line arguments, and environment variables.

Theory

Node.js is a JavaScript runtime built on Chrome's V8 engine that enables server-side scripting. It uses a module system (CommonJS) via `require/module.exports`. The `fs` module provides synchronous and asynchronous file I/O. The `path` and `os` modules give path manipulation and system information. `events.EventEmitter` supports the publish-subscribe pattern. The `http` module creates servers without frameworks. Streams allow efficient reading of large files in chunks. `npm` manages packages; `process.argv` and `process.env` expose CLI arguments and environment variables.

Software Required

- Node.js (v18 or later)
- npm (comes with Node.js)
- Visual Studio Code / any text editor
- Terminal / Command Prompt
- Web browser or curl / Postman (for HTTP routes)

Procedure

1. Create a project folder and write each task's .js file with the given filename.
2. For Task 1, create `sum.js` and `even-odd.js`, import them in `app1.js` and run `node app1.js`.
3. For Tasks 2–6, write and run the respective scripts using the listed Node core modules.
4. For Tasks 7–8, start the HTTP server (`node server.js`) and test routes with browser or curl.
5. For Task 9, run `stream-demo.js` and observe chunk sizes.
6. For Task 10, run `npm init -y`, `npm install nanoid@3`, then `node id-gen.js`.
7. For Tasks 11–12, pass CLI arguments and set `PORT` environment variable as shown.

Source Code

Task 1: Custom Modules

sum.js

```
// sum.js - exports a function that returns the sum of two numbers
function sum(a, b) {
  return a + b;
}
module.exports = sum;
```

even-odd.js

```
// even-odd.js - exports a function that checks even/odd
function evenOdd(n) {
  return n % 2 === 0 ? "Even" : "Odd";
}
module.exports = evenOdd;
```

app1.js

```
// app1.js - imports custom modules and calls them
const sum = require("./sum");
const evenOdd = require("./even-odd");

console.log("Sum of 15 and 27 =", sum(15, 27));
console.log("Number 42 is", evenOdd(42));
console.log("Number 17 is", evenOdd(17));
```

Task 2: File System — CRUD

file-ops.js

```
// file-ops.js - CRUD operations on a text file
const fs = require("fs");

// Create + Write
fs.writeFileSync("student.txt", "Name: Utsav Kumar Kushwaha\nRoll No: _____\n");
console.log("1. student.txt created and written.");

// Append
fs.appendFileSync("student.txt", "Course: Full Stack Web Development\n");
console.log("2. Course name appended.");

// Read
const content = fs.readFileSync("student.txt", "utf8");
console.log("3. File content:\n" + content);

// Rename
fs.renameSync("student.txt", "profile.txt");
console.log("4. File renamed to profile.txt");
```

Task 3: Directory Operations

dir-ops.js

```
// dir-ops.js - Directory operations
const fs = require("fs");
const path = require("path");

const dir = "uploads";

// Create folder
if (!fs.existsSync(dir)) {
  fs.mkdirSync(dir);
  console.log("1. Folder 'uploads' created.");
}
```

```
// Create 3 empty files
["file1.txt", "file2.txt", "file3.txt"].forEach((f) => {
  fs.writeFileSync(path.join(dir, f), "");
});
console.log("2. Three empty files created inside uploads/.");

// List files
const files = fs.readdirSync(dir);
console.log("3. Files in uploads/:", files);

// Delete one file
fs.unlinkSync(path.join(dir, "file2.txt"));
console.log("4. file2.txt deleted.");
console.log("5. Remaining files:", fs.readdirSync(dir));
```

Task 4: Path Module

path-demo.js

```
// path-demo.js - Path module demo
const path = require("path");

const filePath = "/home/user/data/report.pdf";

console.log("Directory :", path.dirname(filePath));
console.log("Base name :", path.basename(filePath));
console.log("Extension :", path.extname(filePath));
console.log("Resolved  :", path.resolve("docs", "notes.txt"));
```

Task 5: OS Module Report

sys-report.js

```
// sys-report.js - OS module system report
const os = require("os");

console.log("Platform      :", os.platform());
console.log("Architecture  :", os.arch());
console.log("Total Memory  :", (os.totalmem() / 1024 / 1024 / 1024).toFixed(2), "GB");
console.log("Free Memory   :", (os.freemem() / 1024 / 1024 / 1024).toFixed(2), "GB");
console.log("CPU Cores     :", os.cpus().length);
console.log("Uptime        :", (os.uptime() / 60).toFixed(2), "minutes");
```

Task 6: Events Module — Custom EventEmitter

order-events.js

```
// order-events.js - Custom EventEmitter
const EventEmitter = require("events").EventEmitter;
const orderEmitter = new EventEmitter();

orderEmitter.on("orderPlaced", (order) => {
  console.log("===== RECEIPT =====");
  console.log("Order ID :", order.id);
  console.log("Item      :", order.item);
  console.log("Amount   : ₹" + order.amount);
  console.log("=====\\n");
});

orderEmitter.emit("orderPlaced", { id: 101, item: "Wireless Earbuds", amount: 2499 });
orderEmitter.emit("orderPlaced", { id: 102, item: "Smart Watch X5", amount: 4999 });
orderEmitter.emit("orderPlaced", { id: 103, item: "USB-C Hub", amount: 2199 });
```

Task 7 & 8: HTTP Server with Manual Routing + Query String

server.js

```
// server.js - HTTP Server with manual routing + query string (Tasks 7 & 8)
const http = require("http");
const url = require("url");
```

```
const students = [
  { id: 1, name: "Utsav Kumar", course: "CSE" },
  { id: 2, name: "Ananya Sharma", course: "CSE" },
  { id: 3, name: "Rahul Verma", course: "IT" }
];

const server = http.createServer((req, res) => {
  const parsed = url.parse(req.url, true);
  const pathname = parsed.pathname;

  // GET /
  if (pathname === "/" && req.method === "GET") {
    res.writeHead(200, { "Content-Type": "text/plain" });
    res.end("Welcome to the Node.js Student API!");
    return;
  }

  // GET /students
  if (pathname === "/students" && req.method === "GET") {
    res.writeHead(200, { "Content-Type": "application/json" });
    res.end(JSON.stringify(students, null, 2));
    return;
  }

  // GET /students/:id
  const studentMatch = pathname.match(/^\/students\/(\d+)$/);
  if (studentMatch && req.method === "GET") {
    const id = parseInt(studentMatch[1], 10);
    const student = students.find((s) => s.id === id);
    if (student) {
      res.writeHead(200, { "Content-Type": "application/json" });
      res.end(JSON.stringify(student, null, 2));
    } else {
      res.writeHead(404, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ error: "Student not found" }));
    }
    return;
  }

  // GET /search?keyword=...
  if (pathname === "/search" && req.method === "GET") {
    const keyword = parsed.query.keyword || "(none)";
    res.writeHead(200, { "Content-Type": "text/plain" });
    res.end("Search keyword received: " + keyword);
    return;
  }

  // 404 for everything else
  res.writeHead(404, { "Content-Type": "text/plain" });
  res.end("404 - Page Not Found");
});

const PORT = 3000;
server.listen(PORT, () => {
  console.log("Server running at http://localhost:" + PORT);
});
```

Task 9: Reading a Large File with Streams

stream-demo.js

```
// stream-demo.js - Reading a large file with streams
const fs = require("fs");

// Create a file with 50+ lines
let content = "";
for (let i = 1; i <= 60; i++) {
  content += "Line " + i + ": This is sample data for stream demonstration.\n";
}
fs.writeFileSync("large-file.txt", content);
console.log("large-file.txt created with 60 lines.\n");

// Read using stream
const readStream = fs.createReadStream("large-file.txt", { encoding: "utf8", highWaterMark: 64 });

readStream.on("data", (chunk) => {
  console.log("Chunk received - size:", Buffer.byteLength(chunk), "bytes");
});

readStream.on("end", () => {
  console.log("\nStream ended - entire file read.");
});

readStream.on("error", (err) => {
  console.error("Error:", err.message);
});
```

Task 10: npm and External Package (nanoid)

id-gen.js

```
// id-gen.js - Generate unique IDs using nanoid
const { nanoid } = require("nanoid");

console.log("Generated 5 unique IDs:");
for (let i = 1; i <= 5; i++) {
  console.log(i + ". ", nanoid());
}
```

Commands: npm init -y then npm install nanoid@3

Task 11: Command-Line Arguments

calc-cli.js

```
// calc-cli.js - Command-line calculator
const args = process.argv.slice(2);

if (args.length < 3) {
  console.log("Usage: node calc-cli.js <num1> <num2> <operator>");
  console.log("Example: node calc-cli.js 10 5 +");
  process.exit(1);
}

const a = parseFloat(args[0]);
const b = parseFloat(args[1]);
const op = args[2];

let result;
switch (op) {
  case "+": result = a + b; break;
  case "-": result = a - b; break;
  case "*": result = a * b; break;
  case "/": result = b !== 0 ? a / b : "Error: Division by zero"; break;
  default:
    console.log("Unsupported operator. Use +, -, *, /");
}
```

```
    process.exit(1);
  }

  console.log(`${a} ${op} ${b} = ${result}`);
```

Task 12: Environment Variables

env-demo.js

```
// env-demo.js - Environment variables
const http = require("http");

const PORT = process.env.PORT || 4000;

const server = http.createServer((req, res) => {
  res.writeHead(200, { "Content-Type": "text/plain" });
  res.end("Server is running on port " + PORT);
});

server.listen(PORT, () => {
  console.log("Active port:", PORT);
  console.log("Server started at http://localhost:" + PORT);
});
```

Output (Terminal)

Terminal outputs obtained after running each program:

TASK 1 – Custom Modules (node app1.js)

Sum of 15 and 27 = 42

Number 42 is Even

Number 17 is Odd

TASK 2 – File System CRUD (node file-ops.js)

1. student.txt created and written.

2. Course name appended.

3. File content:

Name: Utsav Kumar Kushwaha

Roll No: _____

Course: Full Stack Web Development

4. File renamed to profile.txt

TASK 3 – Directory Operations (node dir-ops.js)

1. Folder 'uploads' created.

2. Three empty files created inside uploads/.

3. Files in uploads/: ['file1.txt', 'file2.txt', 'file3.txt']

4. file2.txt deleted.

5. Remaining files: ['file1.txt', 'file3.txt']

TASK 4 – Path Module (node path-demo.js)

Directory : /home/user/data

Base name : report.pdf

Extension : .pdf

Resolved : /home/workdir/artifacts/nodejs-lab/docs/notes.txt

TASK 5 – OS Module Report (node sys-report.js)

Platform : linux

Architecture : x64

Total Memory : 1.20 GB

Free Memory : 0.91 GB

CPU Cores : 2

Uptime : 2.30 minutes

TASK 6 – Events Module (node order-events.js)

===== RECEIPT =====

Order ID : 101

Item : Wireless Earbuds

Amount : ₹2499

=====

===== RECEIPT =====

Order ID : 102

Item : Smart Watch X5

Amount : ₹4999

=====

===== RECEIPT =====

Order ID : 103

Item : USB-C Hub

Amount : ₹2199

=====

TASK 7 & 8 – HTTP Server + Query String

=== GET / ===

Welcome to the Node.js Student API!

=== GET /students ===

```
[
  {
    "id": 1,
    "name": "Utsav Kumar",
    "course": "CSE"
  },
```

```
{
  "id": 2,
  "name": "Ananya Sharma",
  "course": "CSE"
},
{
  "id": 3,
  "name": "Rahul Verma",
  "course": "IT"
}
]
=== GET /students/2 ===
{
  "id": 2,
  "name": "Ananya Sharma",
  "course": "CSE"
}
=== GET /students/99 ===
{"error": "Student not found"}
=== GET /search?keyword=node ===
Search keyword received: node
=== GET /unknown ===
404 - Page Not Found
```

TASK 9 - Streams (node stream-demo.js)
large-file.txt created with 60 lines.

[illegible]

```
Chunk received - size: 64 bytes
Chunk received - size: 64 bytes
Chunk received - size: 64 bytes
Chunk received - size: 64 bytes
Chunk received - size: 27 bytes
```

```
Stream ended - entire file read.
```

```
TASK 10 - nanoid (node id-gen.js)
Generated 5 unique IDs:
1. GzDThfJAQ126rwwGV2COH
2. mOIJyZ7XbEEjj5aIRIdVa
3. GaYDNq_bck_F34Hsc0az6
4. fGi8LiZ8Ix3amNSqd41BU
5. WZeHAWaqrwtZ0M8KdG5e
```

```
TASK 11 - CLI Calculator
10 + 5 = 15
```

```
20 / 4 = 5
```

```
TASK 12 - Environment Variables
Active port: 5000
Server started at http://localhost:5000
```

Result

All twelve Node.js tasks were successfully implemented and executed. Custom modules were imported and used; file and directory CRUD operations worked correctly; path and OS information were printed; EventEmitter produced formatted receipts; the HTTP server responded correctly to all routes including query strings; streams read the large file in chunks; nanoid generated unique IDs; the CLI calculator computed results from process.argv; and the server started on a PORT taken from process.env. Core Node.js concepts required by Lab Sheet 4 have been demonstrated.

-OUTPUT

Task 1: Custom Modules (sum.js, even-odd.js, app1.js) <pre>PS C:\lab> node app1.js Sum of 10 and 15 is: 25 Is 10 even? true Is 7 odd? true Is 8 even? true Is 9 odd? true</pre>	Task 2: File System — CRUD on a Text File (file-ops.js) <pre>PS C:\lab> node file-ops.js File created: student.txt Name and Roll No written. Course appended. ----- Full content of file: Name: Utsav Kumar Kushwaha Roll No: 210123 Course: Full Stack Web Development File renamed to: profile.txt</pre>	Task 3: Directory Operations (dir-ops.js) <pre>PS C:\lab> node dir-ops.js Folder 'uploads' created. 3 files created inside 'uploads'. Files in uploads folder: 1. file2.txt 2. file2.txt 3. file3.txt Deleted file2.txt Remaining files: 1. file1.txt 2. file3.txt</pre>	Task 4: Path Module (path-demo.js) <pre>PS C:\lab> node path-demo.js File path: /home/user/data/report.pdf Directory name: /home/user/data Base name: report.pdf Extension: .pdf Resolved absolute path: C:\lab\data\report.pdf</pre>
Task 5: OS Module Report (sys-report.js) <pre>PS C:\lab> node sys-report.js Platform: win32 Architecture: x64 Total Memory: 16329.56 MB Free Memory: 8123.45 MB CPU Cores: 8 Uptime: 125.32 minutes</pre>	Task 6: Events Module — Custom EventEmitter (order-events.js) <pre>PS C:\lab> node order-events.js Order Placed! ID: ORD001 Item: Laptop Amount: ₹45000 ----- Order Placed! ID: ORD002 Item: Mouse Amount: ₹1200 ----- Order Placed! ID: ORD003 Item: Keyboard Amount: ₹2500</pre>	Task 7: HTTP Server with Manual Routing (server.js) <pre>PS C:\lab> node server.js Server running at http://localhost:3000 Tested Routes (in browser/Postman): GET / → Welcome to our Student Portal! GET /students [{ "id": 1, "name": "Aman", "course": "CSE"}, { "id": 2, "name": "Neha", "course": "ECE"}, { "id": 3, "name": "Rohit", "course": "ME" }] GET /students/2 → { "id": 2, "name": "Neha", "course": "ECE" } GET /random → 404 Not Found (JSON)</pre>	Task 8: Query String Parsing (server.js / search route) <pre>PS C:\lab> curl "http://localhost:3000/search?keyword=node" { "keyword": "node" }</pre>
Task 9: Reading a Large File with Streams (stream-demo.js) <pre>PS C:\lab> node stream-demo.js Created large.txt with 50 lines. Reading file using createReadStream... Received chunk of size: 1024 bytes Received chunk of size: 1024 bytes Received chunk of size: 1024 bytes Received chunk of size: 1024 bytes Received chunk of size: 856 bytes File reading completed.</pre>	Task 10: npm and an External Package (id-gen.js) <pre>PS C:\lab> npm init -y Wrote to C:\lab\package.json: { "name": "lab4", "version": "1.0.0", "main": "index.js", "license": "ISC" } PS C:\lab> npm install nanoid@3 added 1 package, and audited 1 package in 2s found 0 vulnerabilities</pre> <pre>PS C:\lab> node id-gen.js Generated IDs: 1. Vj8fk2ZqQe1 2. 9kL3mN7pT2X 3. b67dYvH9G9J 4. Zf4tLm8nP3Q 5. Rp2xC9vKd6M</pre>	Task 11: Command-Line Arguments (calc-cli.js) <pre>PS C:\lab> node calc-cli.js 10 5 + Result: 15 PS C:\lab> node calc-cli.js 10 5 - Result: 5 PS C:\lab> node calc-cli.js 10 5 * Result: 50 PS C:\lab> node calc-cli.js 10 5 / Result: 2</pre>	Task 12: Environment Variables (env-demo.js) (Windows) <pre>PS C:\lab> set PORT=5000 && node env-demo.js Server running on port 5000</pre> (Linux / Mac) <pre>\$ PORT=5000 node env-demo.js Server running on port 5000</pre>